

Pranav Rathod

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EDUCATION

University of Southern California

Master of Science in Computer Science, Multimedia and Creative Technologies (GPA: 3.68/4.00)

January 2024 – Present

Los Angeles, CA

University of Illinois at Chicago

Bachelor of Science in Computer Science, Minor in Art (GPA: 3.92/4.00)

August 2019 – May 2023

Chicago, IL

RELEVANT COURSEWORK

Multimedia Systems Design, Analysis of Algorithms, Computer Graphics, Computer Animation and Simulation, Game Engine Development, 3D Graphics and Rendering, Augmented/Virtual Reality, Artificial Intelligence, Framework-based Development

SKILLS

Languages: C/C++, Python, C#, Java, JavaScript, SQL, Dart, Go, F#

Technical: OOP, Git, CI/CD (GitHub Actions), Docker, Terraform, Linux/Unix, OpenGL, PyTorch, Node.js, React, HTML5, CSS3, Unity, WebGL, Firebase, GCP, SQLite3, Multi-Threading, TCP/IP, Testing Frameworks (Google/JUnit5)

EXPERIENCE

Media Support Specialist

Annenberg TechOps Information Technology, University of Southern California

Los Angeles, CA

January 2024 – Present

- Resolving asset pipeline issues, and debugging cross-platform rendering problems to support deploying multimedia projects
- Answering technical queries involving with HTML, Adobe Creative Cloud and WordPress for students and faculty
- Producing screen casts and step-by-step tutorials to aid users in multimedia production workflows

Research Assistant

ELiCIT Lab, University of Illinois at Chicago

Chicago, IL

June 2022 – December 2023

- Collaborated with Adobe Research to prototype a Flutter-based mobile app for multimodal interaction studies
- Developed UI workflows and backend logic to manage speech and gesture-driven image editing tasks across 12 workflows
- Integrated GCP Speech-to-Text API and installed real-time gesture logging using touch coordinates and timing data
- Explored backend trade-offs in storing synchronized multimodal data for use in training NUI models

Teaching Assistant

Department of Computer Science, University of Illinois at Chicago

Chicago, IL

January 2022 – December 2022

- Guided students in debugging and writing code in C, C++ and building applications using Flutter, Dart across 3 semesters
- Formulated course logistics and graded assignments for intermediate and advanced Computer Science courses each semester
- Managed community discussions and provided instructional support to more than 100 students at once

PROJECTS

PrimeEngine | C++, OpenGL, Maya

April 2025

- Reduced debugging overhead and observability by building a real-time 2D UI overlay system in a custom C++ game engine
- Improved processing efficiency by 40% via AABB-based frustum culling and visual debugging tools for bounding volumes
- Integrated Maya asset workflow and optimized render loop to support faster design-to-engine iteration cycles

Motion Capture Interpolation | C++, OpenGL, FLTK

March 2025

- Reconstructed dropped motion capture sequences using Linear, Bezier, SLERP, and Bezier SLERP interpolation methods
- Programmed custom GLSL shaders with Phong shading to visualize interpolated joint rotations in real time
- Built OOP-based playback system with FLTK UI to compare interpolation quality in real time

Segmented Video Compression | Python, Detectron2, OpenCV, PyTorch

December 2024

- Designed a compression pipeline to encode .rgb videos by applying DCT, quantization, and motion estimation
- Advanced segmentation by using Detectron2 and optical flow, achieving 95% accuracy in foreground-background separation
- Constructed a video player enabling 30 FPS playback with real-time IDCT reconstruction, and synchronized audio

Traffic Crash Analysis Using ML | Python, Pandas, JavaScript

Apr 2023

- Engineered a data pipeline to cluster and classify 1M+ city-wide traffic crash records leveraging K-Means and Naive Bayes
- Extracted key features by region, vehicle type, and seasonality to improve interpretability and model performance
- Strengthened machine learning fundamentals in preprocessing, model training, and metric-driven evaluation

ACHIEVEMENTS & LEADERSHIP

SIGGRAPH Student Volunteer Team Leader | Denver, CO; Tokyo, Japan; Vancouver, BC

2024–2025

- Oversaw operations for 350+ international student volunteers at world's premier computer graphics conference
- Facilitated event logistics and attendee support, collaborating with global teams of students and industry professionals

Undergraduate Research Forum | Chicago, IL

April 2023

- Achieved 3rd place in the Engineering and Physical Sciences category for presenting NUI-based research

Engineering Expo | Chicago, IL

April 2021

- Won 'Best in Category' for designing an Arduino-based device to enforce COVID-19 social distancing